

AGILE WORK

Project Management (PMAE/PM)

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Learning objectives

- Understand the basic principles of agile working
- Apply principles of agile working
- Frameworks and approaches to agile
Know software development
- Understand iterative product development and
can implement
- How SCRUM works as an agile approach
Understanding the working methods



AGILE PROJECT CULTURE

Description of agile development

- **An agile way of working enables companies to develop better products faster. This makes More satisfied employees and customers.**
- Many large companies such as Miele and C&A have therefore already partially or even completely switched to Agile. Many others are currently planning to introduce agile working methods into their organizations.
- The biggest challenge is the cultural shift within the company. This was stated by 70 percent of respondents in a McKinsey study.

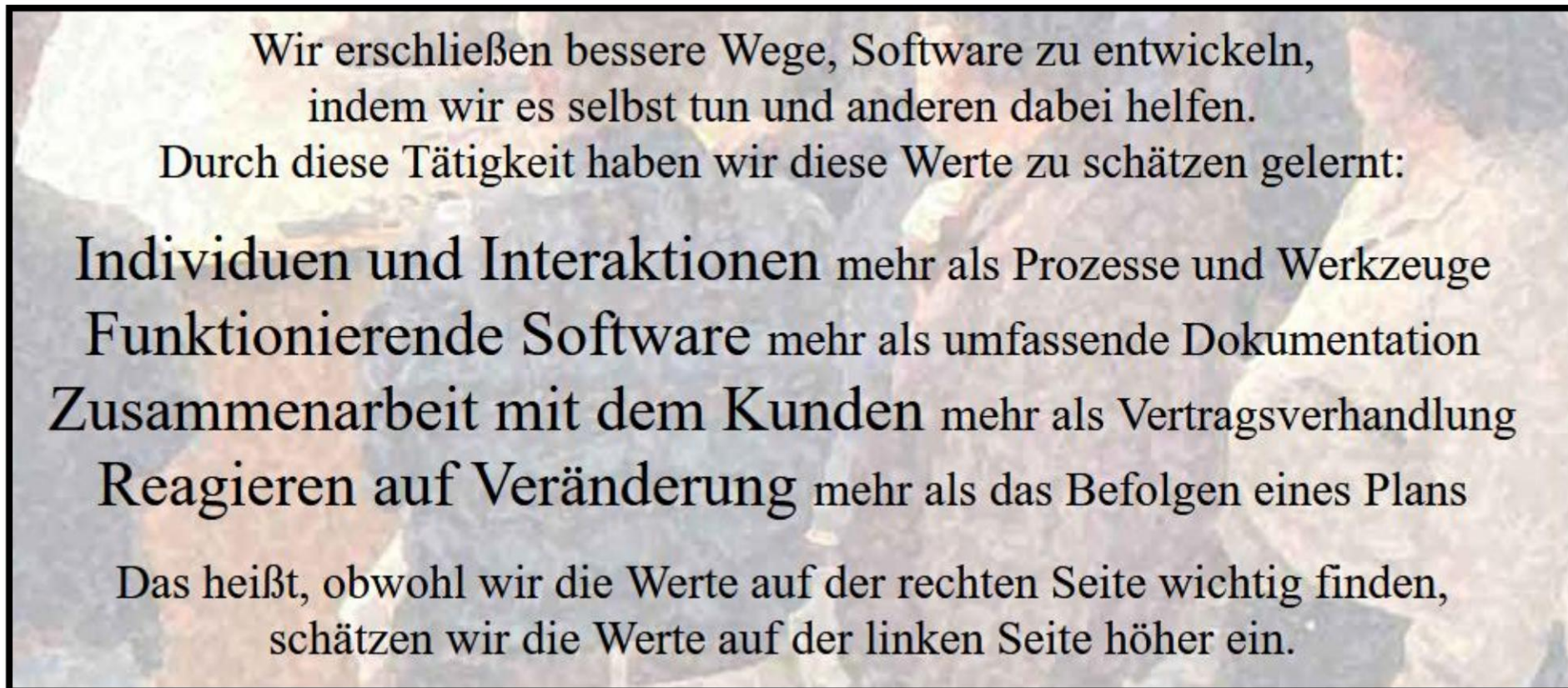
They don't know how to formulate an agile culture and bring it to life.

It is therefore no wonder that very few companies manage to introduce an agile working culture. This was the finding of the 14th State of Agile Report 2020, published annually.

(Source: echometerapp.com – Agile Culture)

Agile Manifesto

The 4 principles



(Source: Agile Manifesto on agilemanifesto.org,
signed by hundreds of people worldwide)

Agile Manifesto

The 12 principles

- Our highest priority is to provide customers with valuable software through early and **continuous delivery**.
to satisfy.
- We welcome even the most challenging **changes in requirements late in development** . Utilize agile processes.
Changes that benefit the customer's competitive advantage.
- Deliver **working software** regularly **within a few weeks or months** , and preferentially...
the shorter time period.
- Subject matter experts and developers must work together daily throughout the project .

(Source: Agile Manifesto on agilemanifesto.org, signed by
hundreds of people worldwide, quoted verbatim)

Agile Manifesto

The 12 principles

- Build projects around **motivated individuals**. Give them the environment and support they need.
I need them and **trust** that they will complete the task.
- The most efficient and effective method of communicating information to and within a development team
Communicating is **done in face-to-face conversation**.
- Working software is the **most important measure of progress**.
- Agile processes promote sustainable development. Clients, developers, and users should have a
be able to maintain **a constant pace for an unlimited time** .

(Source: Agile Manifesto on agilemanifesto.org,
signed by hundreds of people worldwide, quoted verbatim)

Agile Manifesto

The 12 principles

- Constant **focus on technical excellence and Good design** promotes agility.
- **Simplicity** -- the art of maximizing the amount of work not done -- **is essential**.
- The best architectures, requirements, and designs emerge from **self-organized teams**.
- At **regular intervals, the team reflects** on how it can become more effective and adjusts its behavior accordingly.

(Source: Agile Manifesto on agilemanifesto.org, signed by hundreds of people worldwide, quoted verbatim)

Agile approach in comparison

Non-Agile (Classic, e.g., V-model / Waterfall model)	Agil (z.b. SCRUM)
Requirements known at the beginning ; changing requirements during the project is difficult.	Requirements initially unclear
High costs for late requirement changes	Moderate costs for late requirement changes
Requirements description from a technical perspective (features)	Requirements description from the customer's perspective (Use cases)
The customer only sees the final result.	Customer evaluates interim results
When time gets tight, it's better to postpone milestones.	When time is tight, reduce effort.
Clear hierarchy	Self-organized teams
The team is spread across multiple projects.	The team sits together and focuses on a project.
Assign tasks from above	Take on tasks independently
Lots of communication via documents and long meetings	Lots of informal communication and stand-up meetings
Effort estimation by project manager or experts	Effort estimation together as a team

Tasks in agile implementation

The classic project manager...	The agile project manager...
Has a project and task focus	Has a team and process focus
Ensure a clear project mandate, develop a project plan, and manage the project according to the plan.	Create the framework for effective teamwork.
acts as an interface between the departments and the project team (= central Contact person)	Supports the Product Owner, but does not coordinate with the specialist departments .
Is responsible for the results achieved by the team	Is responsible for the productivity of the project team (= trainer)



(Source:
<https://freepik.com>)



(Source:
<https://webstockreview.net>)

THE AGILE WAY TO THE GOAL

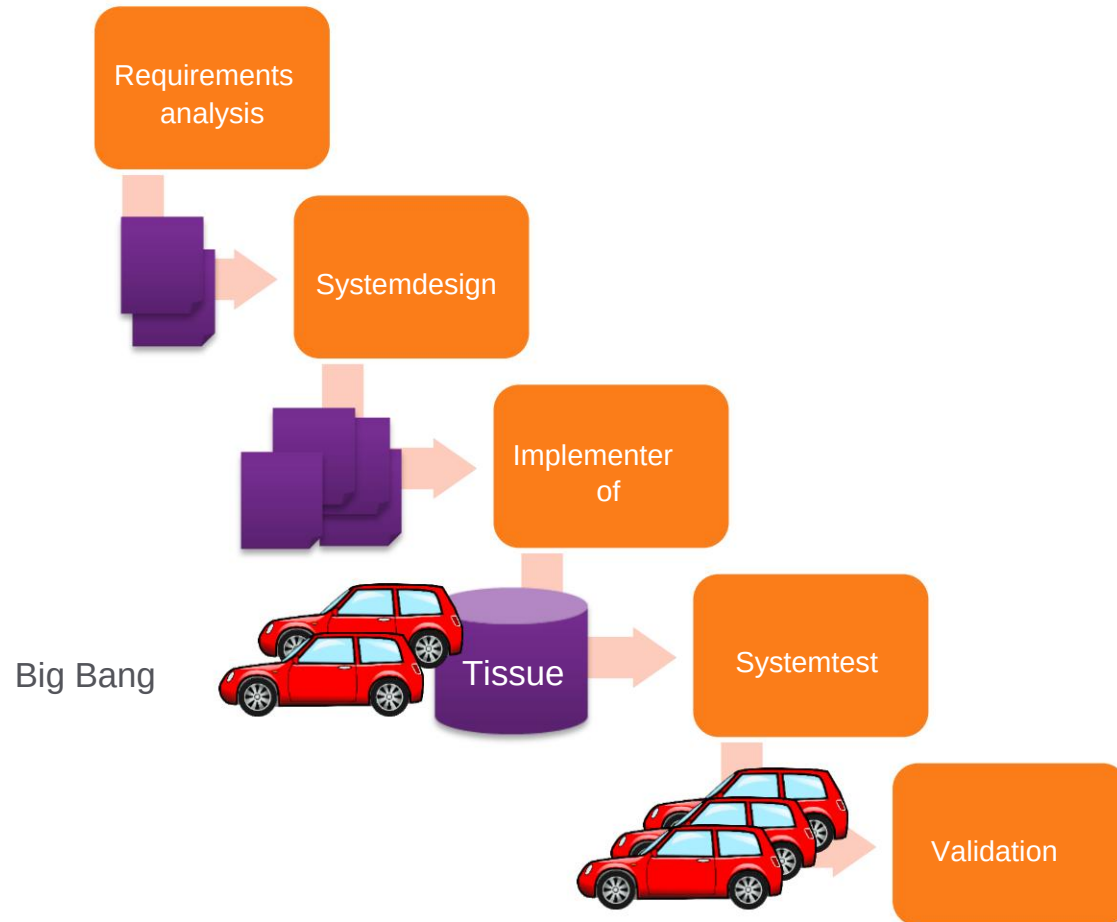
Agile Manifesto (selection)

- Our highest priority is to provide customers with valuable software through early and **continuous delivery** to satisfy.
- Deliver **working software** regularly **within a few weeks or months** , and preferentially... the shorter time period.
- Working software is the **most important measure of progress**.

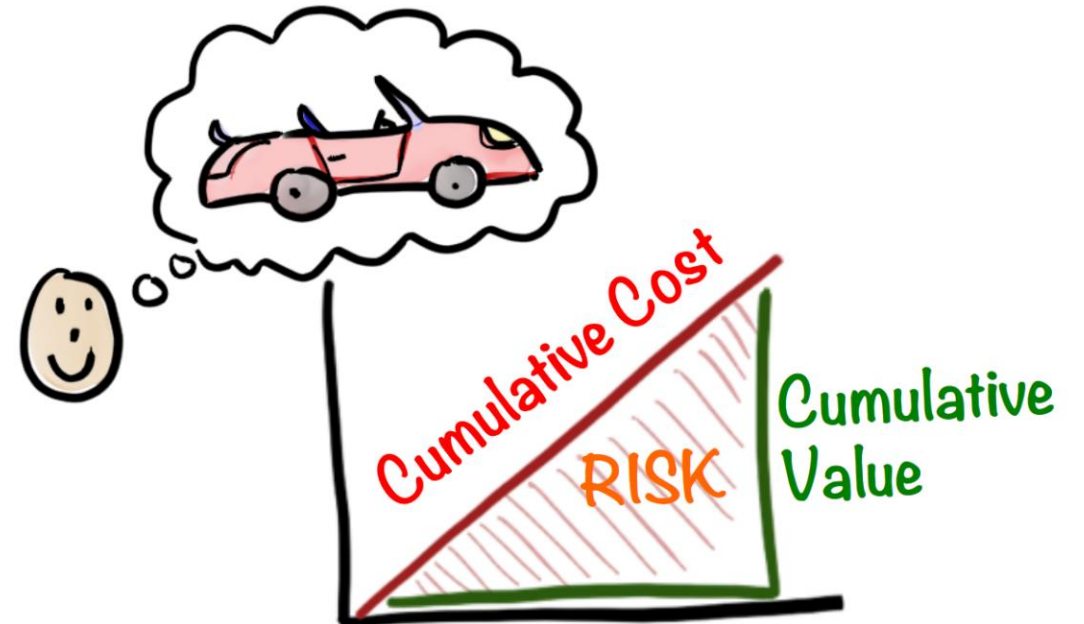


(Source: Agile Manifesto on agilemanifesto.org, signed by hundreds of people worldwide, quoted verbatim)

Development according to the waterfall model



Big Bang = Big Risk



Henrik Kniberg

Goal: To draw a Mona Lisa

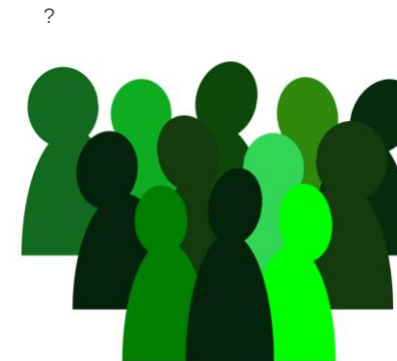
How would you proceed?



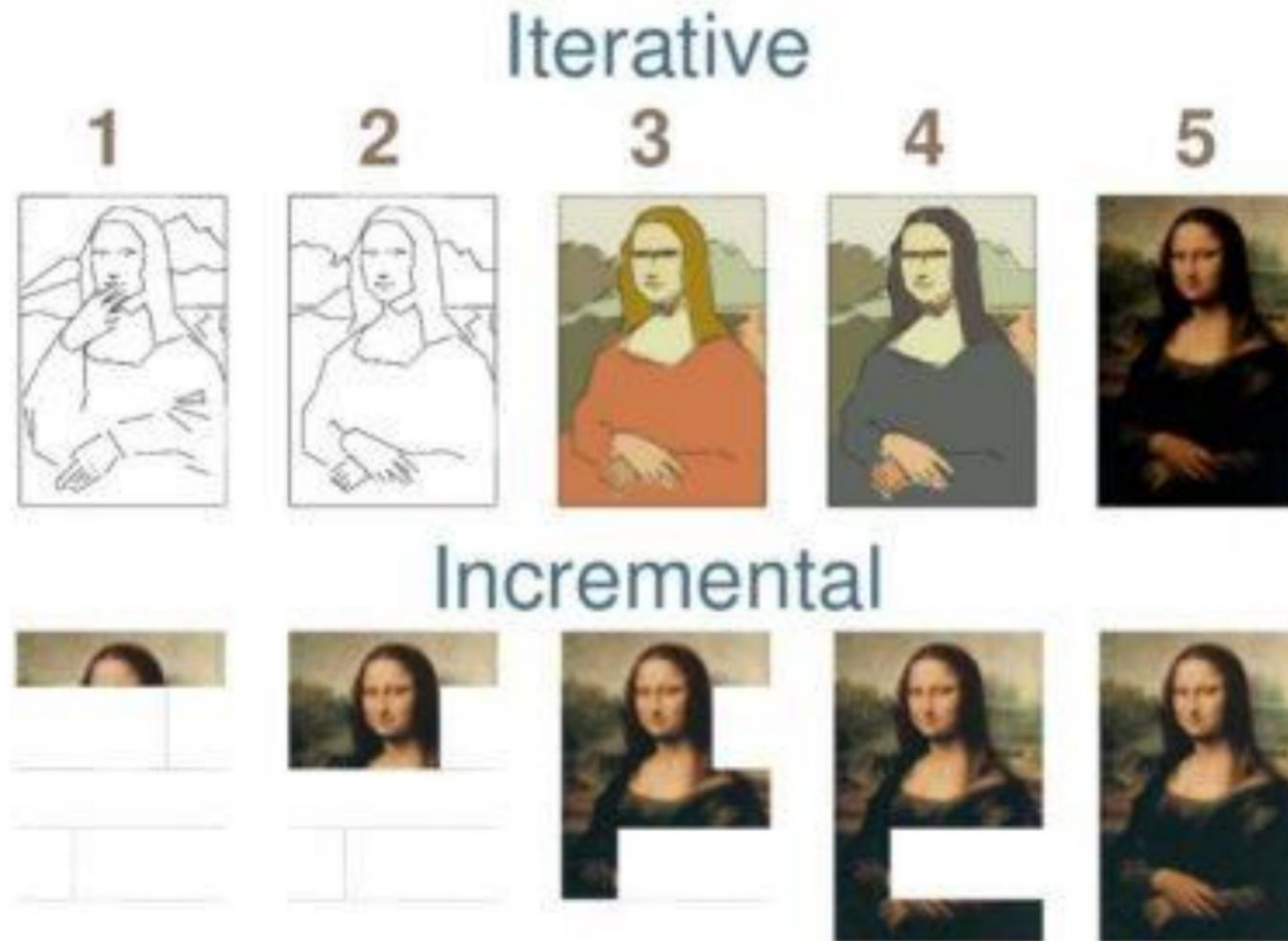
- a) Draw the outlines, then color them in
- b) Divide the image into lines and draw from top to bottom.
- c) First the face, then hair/torso, then background
- d) Define rectangular segments and draw them individually
- e) Draw outlines of important elements, then color them, then draw more outlines, color them again...

“Product” as a measure of work

- Gradual improvement and expansion of a product's functionality
- A functioning product should still be available after a work phase .
- Can be used to interact with product prospects



Iterative vs. Incremental



Which step or procedure will quickly lead to a usable result?

Minimum viable product (MVP)

- Literally: “**minimally usable or viable** product”,
- the **first minimally functional iteration** of a product, which serves to develop from scratch as quickly as possible
Learning from user feedback and thus preventing flawed developments that miss the mark with user requirements.
- In this context, it is important that the iteration offers an initial “**usable**” benefit, so that
that users actually use the product.

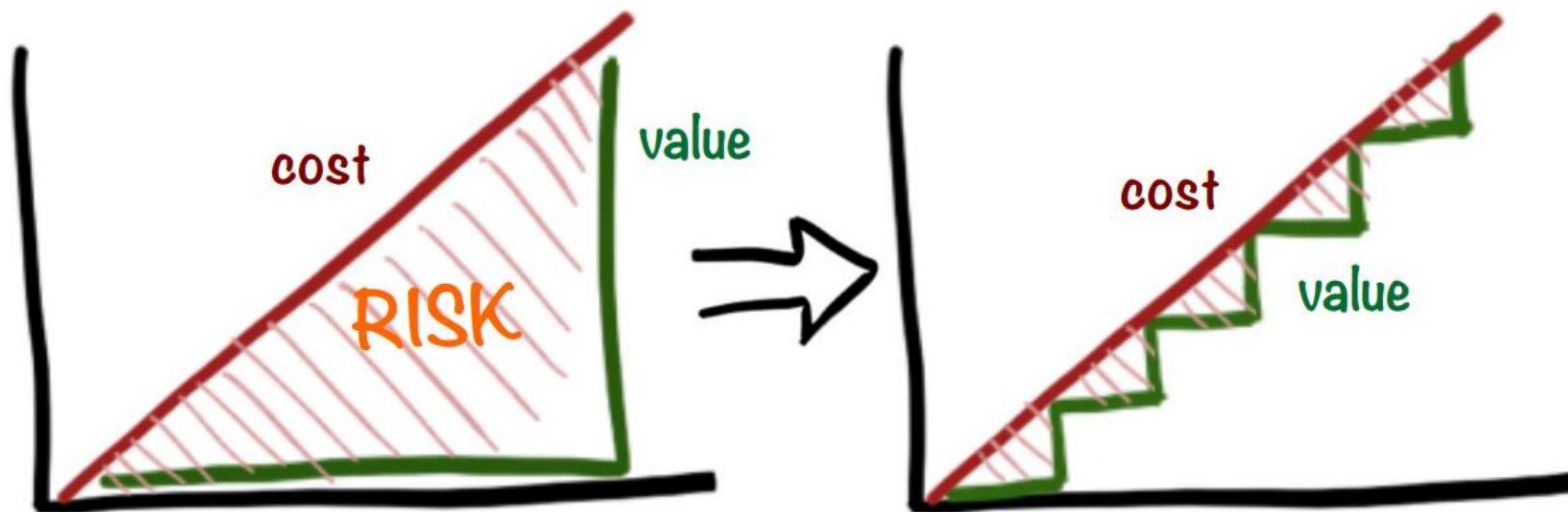
Agile = Iterative + Incremental

Don't try to get it all right
from the beginning

Don't build it all at once

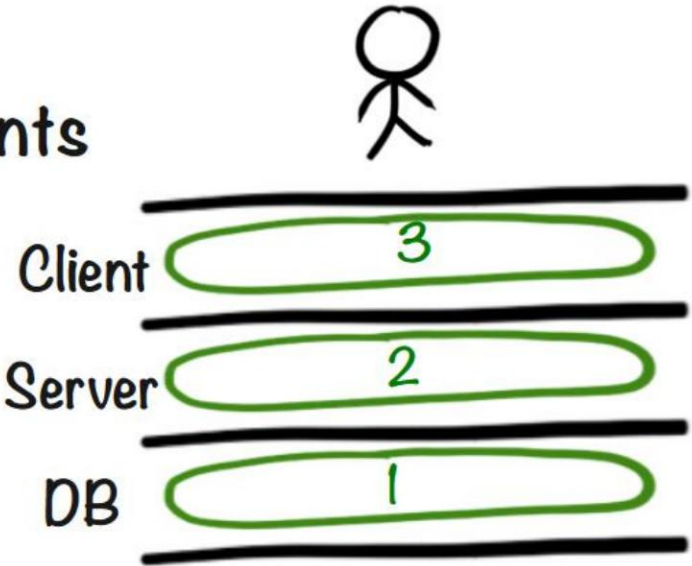
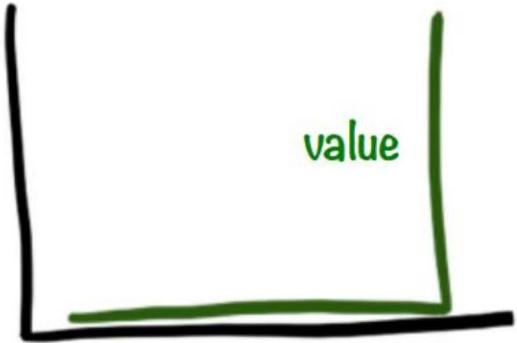


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Amberg-Weiden



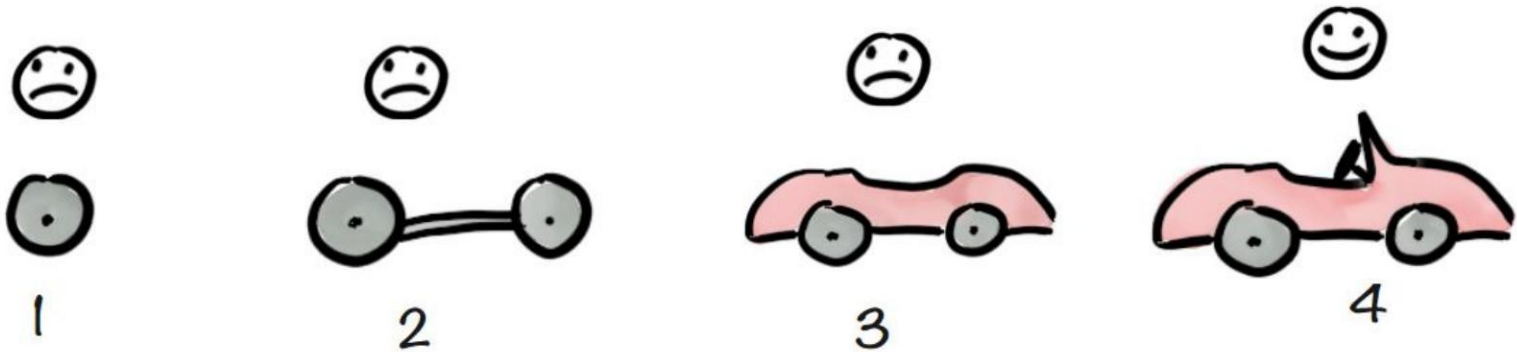
Henrik Kniberg

Not "horizontal" increments



Result

Iteration Nr.

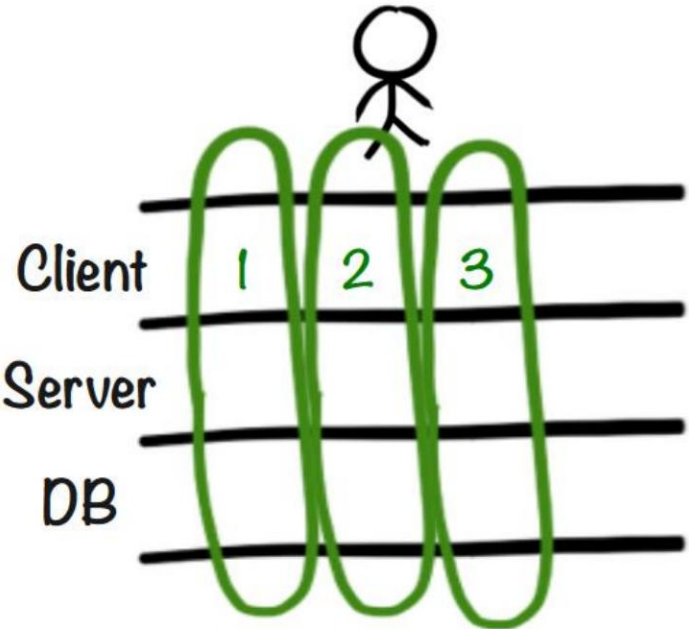
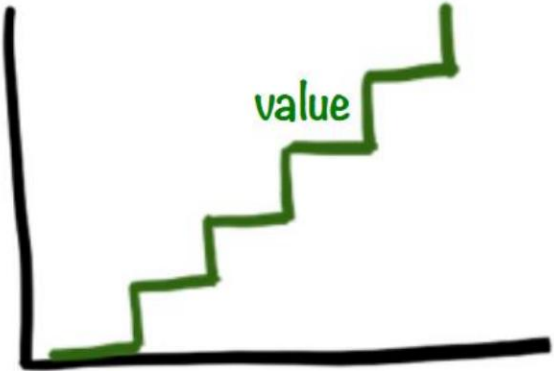


Henrik Kniberg

When will an MVP be available?

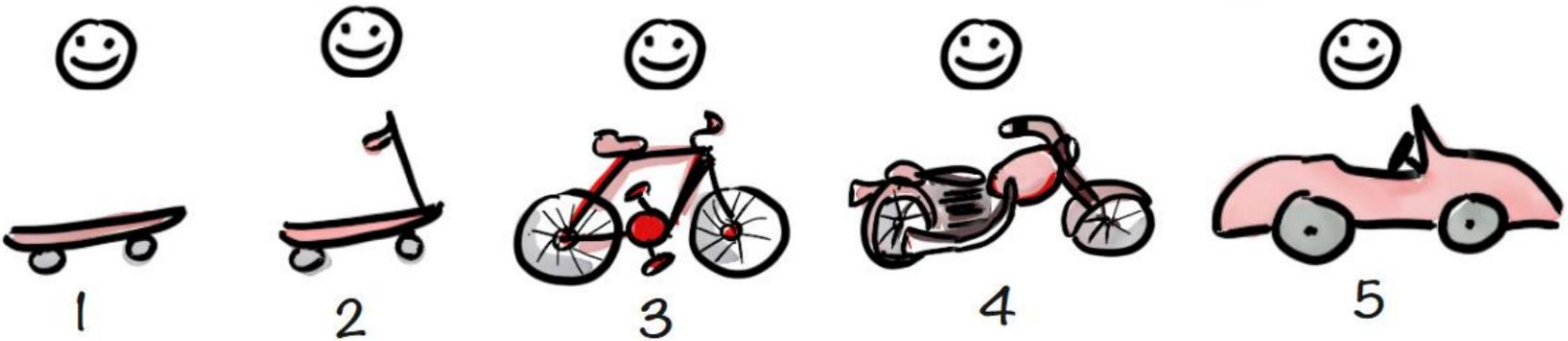
—

“Vertical” increments!



Result

Iteration Nr.



Henrik Kniberg

When will an MVP be available?

Group work

- You are working on a project and planning the construction of a single-family house.
- What is the purpose of the house?
- What would be a possible MVP?
- What would be possible work phases and when would the MVP be available?



Werden Sie
aktiv!

Agile working methods, for example:

House construction

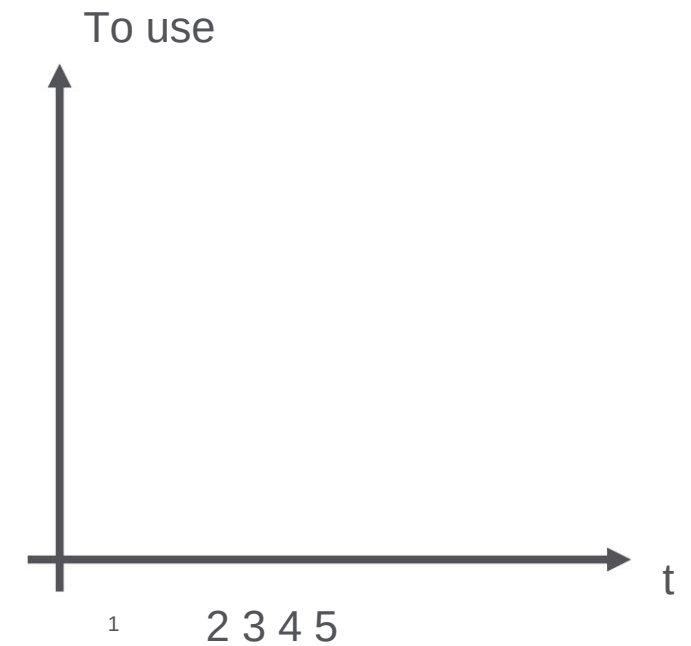
Outline the architecture, necessary work phases, and benefits over time.

architecture



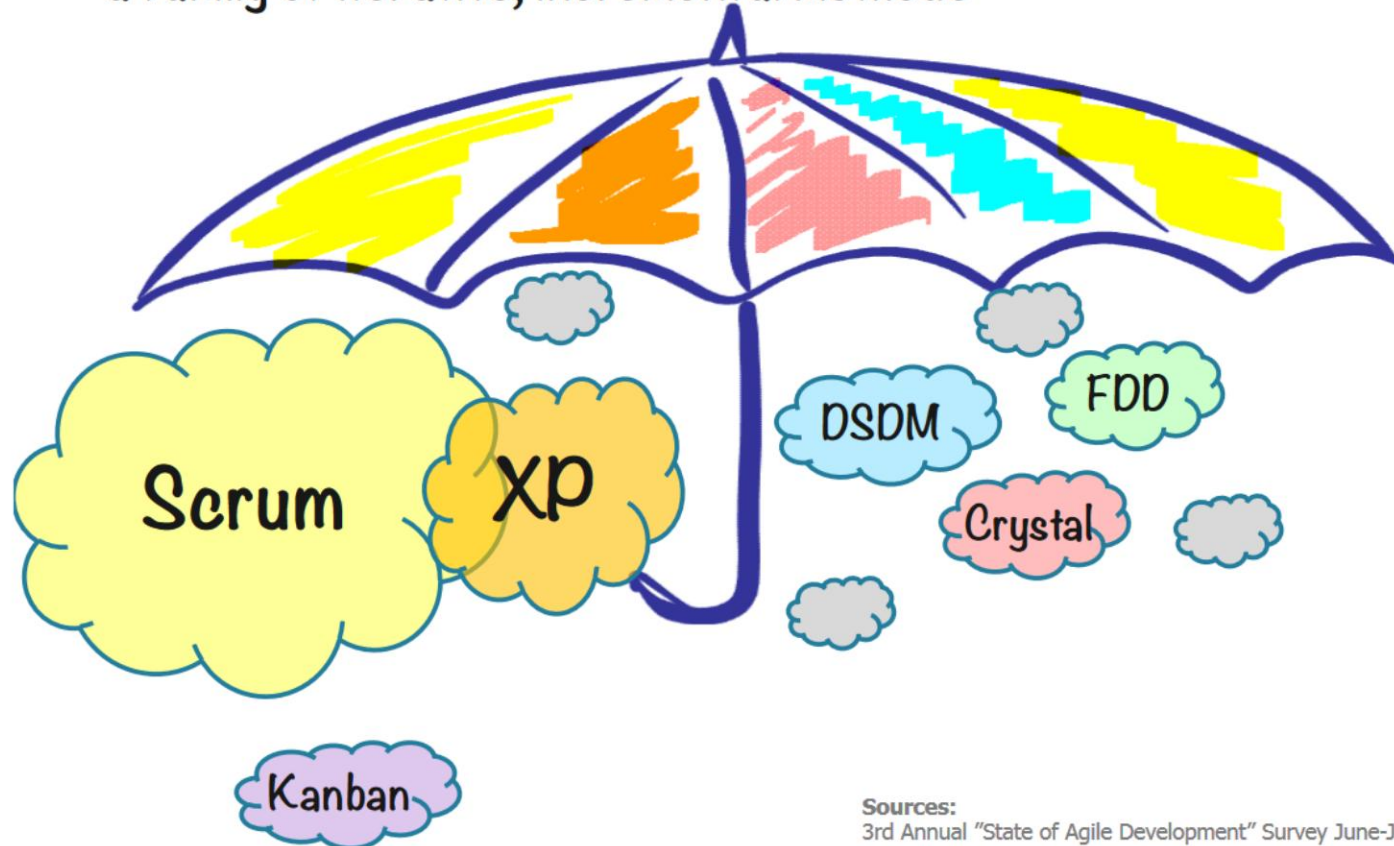
Phases:

- 1.
- 2.
- 3.
- 4.
- 5.



Overview of Agile Methods

Agile "umbrella" –
a family of iterative, incremental methods



Sources:

3rd Annual "State of Agile Development" Survey June-July 2008

- 3061 respondents
- 80 countries

Overview of Agile Methods

- Research one method at a time
- Write down the most important features in bullet points on the digital whiteboard.



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